Algebra Readiness Check · Form A

Name _____ Class _____ Date _____

Choose the best answer and circle its letter. Show any work in the space beside the question. This check is not graded; it helps your teacher see how you are thinking, so a careful guess is better than a blank.

1. Without solving, which equation has the same solution as $3x + 7 = 22$?

   A $3x = 15$
   B $3x = 29$
   C $x + 7 = 19$
   D $3x + 7 = 15$

2. Without adding, decide whether this statement is true or false: $47 + 38 = 48 + 37$.

   A False, because the numbers on the two sides are different.
   B False, because $47 + 38 = 85$, and 85 is not 48.
   C True, because 48 is one more than 47 and 37 is one less than 38.
   D You cannot tell without computing both sides.

3. Which of these is a correct first step for solving $4x - 8 = 12$?

   A $x - 8 = 3$
   B $x - 2 = 3$
   C $4x = 4$
   D $x - 8 = 12$

4. If $x = 6$, what is the value of $4x - x$?

   A 4
   B 18
   C 40
   D 3

5. Which expression is equivalent to $3x + 2 + 5x$?

   A $8x + 2$
   B $10x$
   C $15x + 2$
   D $8x^2 + 2$

6. Which expression is equivalent to $4(x + 3)$?

A $4x + 3$
B $4x + 7$
C $4x + 12$
D $12x$

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7. Which expression is equivalent to $10 - (x - 3)$?

A $13 - x$
B $7 - x$
C $13 + x$
D $x - 13$

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8. Which expression is equivalent to $(x + 5)^2$?

A $x^2 + 25$
B $x^2 + 10x + 25$
C $x^2 + 5x + 25$
D $2x + 10$

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9. Solve $7 - 2x = 15$.

A $x = 4$
B $x = -11$
C $x = 3$
D $x = -4$

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10. Solve $-3x + 4 > 10$.

A $x < -2$
B $x > -2$
C $x > 2$
D $x < -14/3$

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11. What is the value of -3^2?

A 9
B -9
C -6
D 6

12. If $x = -2$, what is the value of $x^2 - 3x$?

A -2
B 2
C -10
D 10

13. Which expression is equivalent to $x^3 \cdot x^4$?

A x^7
B x^{12}
C $12x^2$
D $2x^7$

14. What is the slope of the line through (2, 5) and (6, 13)?

A 2
B 1/2
C -2
D 8

15. What is the y-intercept of the line $y = 3x - 4$?

A 3
B -4
C 4
D 4/3

16. If $f(x) = x^2 - 1$, what is $f(3)$?

A 5
B 4
C $3x^2 - 3$
D 8

17. If $f(x) = 2x + 3$, which expression is equal to $f(x + 1)$?

A $2x + 4$
B $2x + 5$
C $2x^2 + 5x + 3$
D $3x + 4$

18. In which choice is y a function of x?

A $\{(1, 4), (2, 4), (3, 4)\}$
B $\{(4, 1), (4, 2), (4, 3)\}$
C $x = y^2$
D The line $x = 3$

19. At a university there are six times as many students as professors. Let S be the number of students and P the number of professors. Which equation is correct?

A $6S = P$
B $S = P + 6$
C $S = 6P$
D $S \cdot P = 6$

20. Which expression represents 'twice the sum of a number n and 3'?

A $2n + 3$
B $2 + n + 3$
C $2 \cdot n \cdot 3$
D $2(n + 3)$